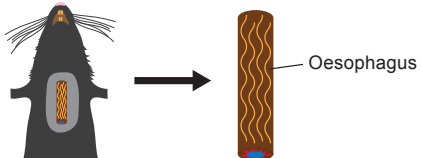
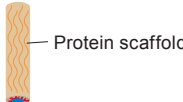
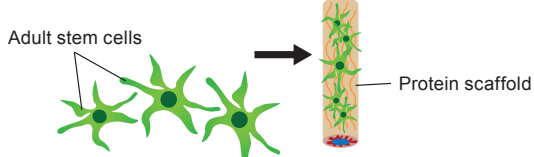
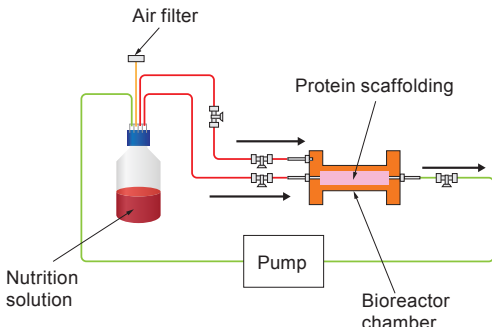
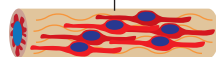


5. In 2018, scientists announced they had produced a laboratory grown oesophagus using a rat oesophagus and human adult stem cells. This was part of on-going research by scientists into different ways of producing organs for transplants.

Image 5 outlines the procedure.

Image 5

Step 1: An oesophagus was removed from a rat	 The diagram shows a rat on the left with a yellow oesophagus highlighted inside its body. An arrow points to the right, where a separate, elongated yellow oesophagus is shown. A label 'Oesophagus' points to this structure.
Step 2: All the rat cells were removed from the oesophagus leaving only a protein scaffold	 The diagram shows a yellow, porous, cylindrical structure representing the protein scaffold. A label 'Protein scaffold' points to it.
Step 3: Human adult stem cells were added to the protein scaffold	 The diagram shows green, star-shaped cells labeled 'Adult stem cells' on the left. An arrow points to the right, where these cells are shown attached to the yellow protein scaffold. A label 'Protein scaffold' points to the structure.
Step 4: The protein scaffold with human adult stem cells was placed in a bioreactor. A bioreactor is a machine which maintains optimum conditions for cell growth	 The diagram shows a bioreactor system. On the left, a bottle of 'Nutrition solution' is connected by a red tube to a 'Pump'. The pump feeds the solution into a 'Bioreactor chamber' which contains the 'Protein scaffolding'. An 'Air filter' is also connected to the top of the chamber. Arrows indicate the flow of the nutrition solution and air into the chamber.
Step 5: At 11 days, the oesophagus was removed from the bioreactor	 The diagram shows the final 'Laboratory grown oesophagus', which is a yellow, porous cylinder with red, star-shaped cells embedded within it. A label 'Laboratory grown oesophagus' points to it.



Examiner
only

(a) (i) State the meaning of the term stem cell. [1]

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(ii) State the name of the type of cell division which occurs during **Step 4**. [1]

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(iii) Suggest **one** advantage of using adult stem cells rather than embryonic stem cells in the process shown in **Image 5**. [1]

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(iv) Suggest **two** variables which need to be controlled in the bioreactor. [2]

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(b) State **one** reason why animal-rights groups might oppose this type of research. [1]

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